CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 10 : Python Program to Implement A* Algorithm

Aim

To develop a Python program that finds the least-cost path from a start node to a goal node using the A* search algorithm.

Algorithm

- Represent the weighted graph as a dictionary and define the heuristic value h(n) for every node.
- Read the start node and the goal node.
- Place the start node in the open list with f(n) = g(n) + h(n).
- Remove the node with the lowest f value from the open list.
- If it is the goal node, print the goal and the total cost, then stop.
- Otherwise mark it visited and add its unvisited neighbours with updated cost and f value.
- If the open list becomes empty, print "Goal not found".

Program

```
graph = {     'A': [('B', 1),
('C', 3)],
    'B': [('D', 2)],
    'C': [('D', 1)],
    'D': [] } h = {'A': 3, 'B': 2,
'C': 1, 'D': 0}

start = input("Enter start node: ")
goal = input("Enter goal node: ")

open_list = [(h[start], 0, start)]
visited = []

while open_list:
    open_list.sort()     f, cost,
node = open_list.pop(0)

    if node == goal:
        print("Goal reached:", node)
print("Cost:", cost)         break

    if node not in visited:
visited.append(node)

        for next_node, distance in graph[node]:             if next_node not in
visited:                 new_cost = cost + distance
open_list.append((new_cost + h[next_node], new_cost, next_node)) else:
    print("Goal not found")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/10.py ==============
Enter start node: A
Enter goal node: D
Goal reached: D
Cost: 3
>>>
```

RESULT

Thus, the A* algorithm was implemented successfully using Python.